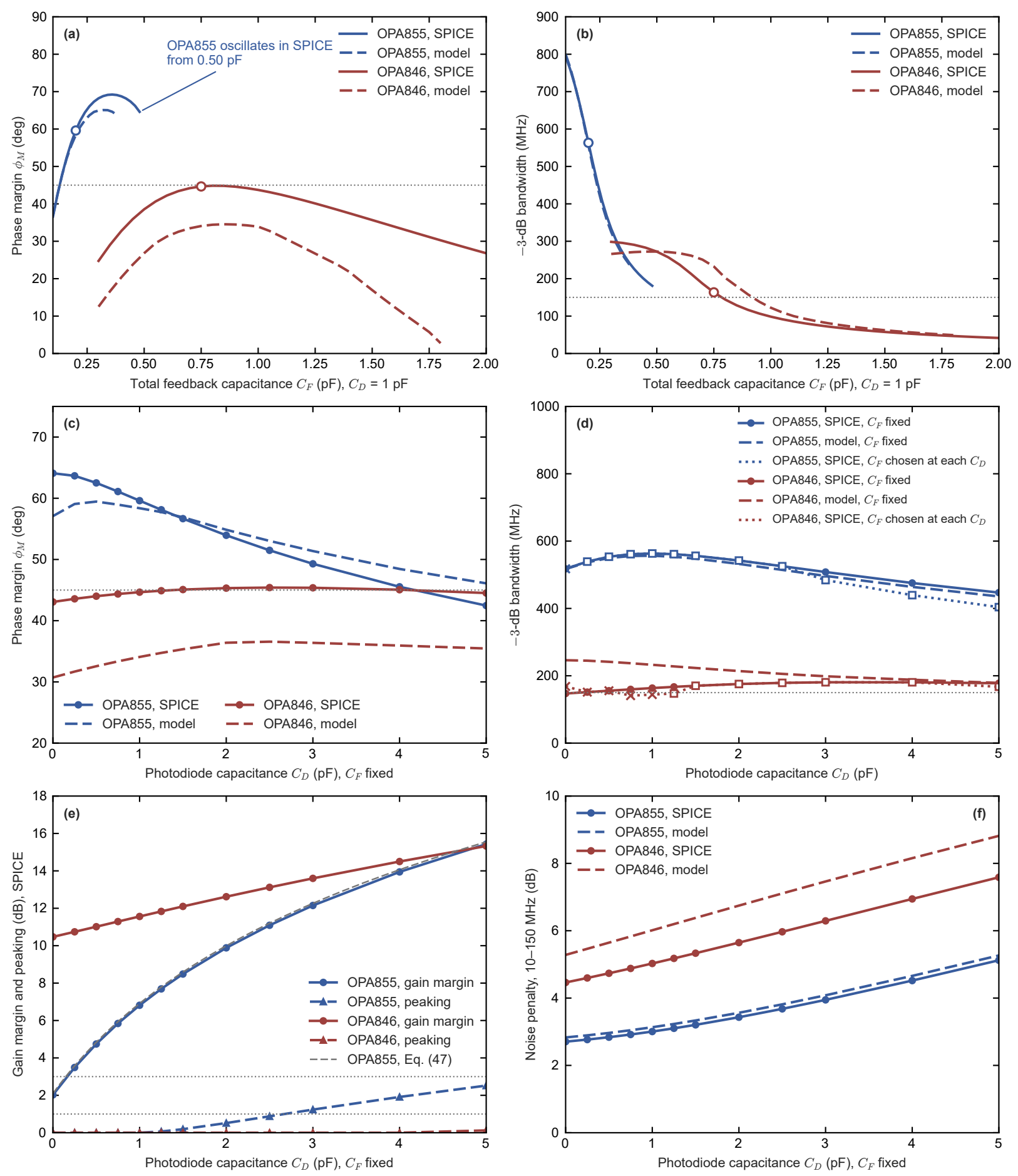

(a)
OPA855 oscillates in SPICE from 0.50 pF
OPA855, SPICE
OPA855, model
OPA846, SPICE
OPA846, model
Phase margin ϕ_M (deg)
Total feedback capacitance C_F (pF), C_D = 1 pF
(b)
OPA855, SPICE
OPA855, model
OPA846, SPICE
OPA846, model
−3-dB bandwidth (MHz)
Total feedback capacitance C_F (pF), C_D = 1 pF
(c)
OPA855, SPICE
OPA855, model
OPA846, SPICE
OPA846, model
Phase margin ϕ_M (deg)
Photodiode capacitance C_D (pF), C_F fixed
(d)
OPA855, SPICE, C_F fixed
OPA855, model, C_F fixed
OPA855, SPICE, C_F chosen at each C_D
OPA846, SPICE, C_F fixed
OPA846, model, C_F fixed
OPA846, SPICE, C_F chosen at each C_D
−3-dB bandwidth (MHz)
Photodiode capacitance C_D (pF)
(e)
OPA855, gain margin
OPA855, peaking
OPA846, gain margin
OPA846, peaking
OPA855, Eq. (47)
Gain margin and peaking (dB), SPICE
Photodiode capacitance C_D (pF), C_F fixed
(f)
OPA855, SPICE
OPA855, model
OPA846, SPICE
OPA846, model
Noise penalty, 10–150 MHz (dB)
Photodiode capacitance C_D (pF), C_F fixed